CSEC PHYSICS

Jan 2023 – Paper 2

Solutions

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SECTION A

Answer ALL questions.

1. (a) Define the term 'period of a simple pendulum' (2 marks)

The period of a simple pendulum is the time taken for the pendulum to complete one full oscillation/cycle, that is for the pendulum to return to the same spot in the same phase.

(b) Table 1 shows the results of an experiment to find the relationship between the length of a string, l, and the period of the pendulum, T.

TABLE 1: LENGTH OF THE STRING, l, AND CORRESPONDING PERIOD VALUES OF A SIMPLE PENDULUM, T.

Length of String, l (m)	Period, T (s)	$Period^2$, T^2 (s^2)
0.15	0.77	0.59
0.25	0.99	0.98
0.35	1.15	1.32
0.45	1.34	1.80
0.55	1.48	2.19
0.65	1.62	2.62
0.75	1.75	3.06

(i) State the independent (manipulated) variable in this experiment.

Length of the string (1 mark)

(ii) Complete Table 1 by inserting the $Period^2$, T^2 values in column 3. (4 marks)

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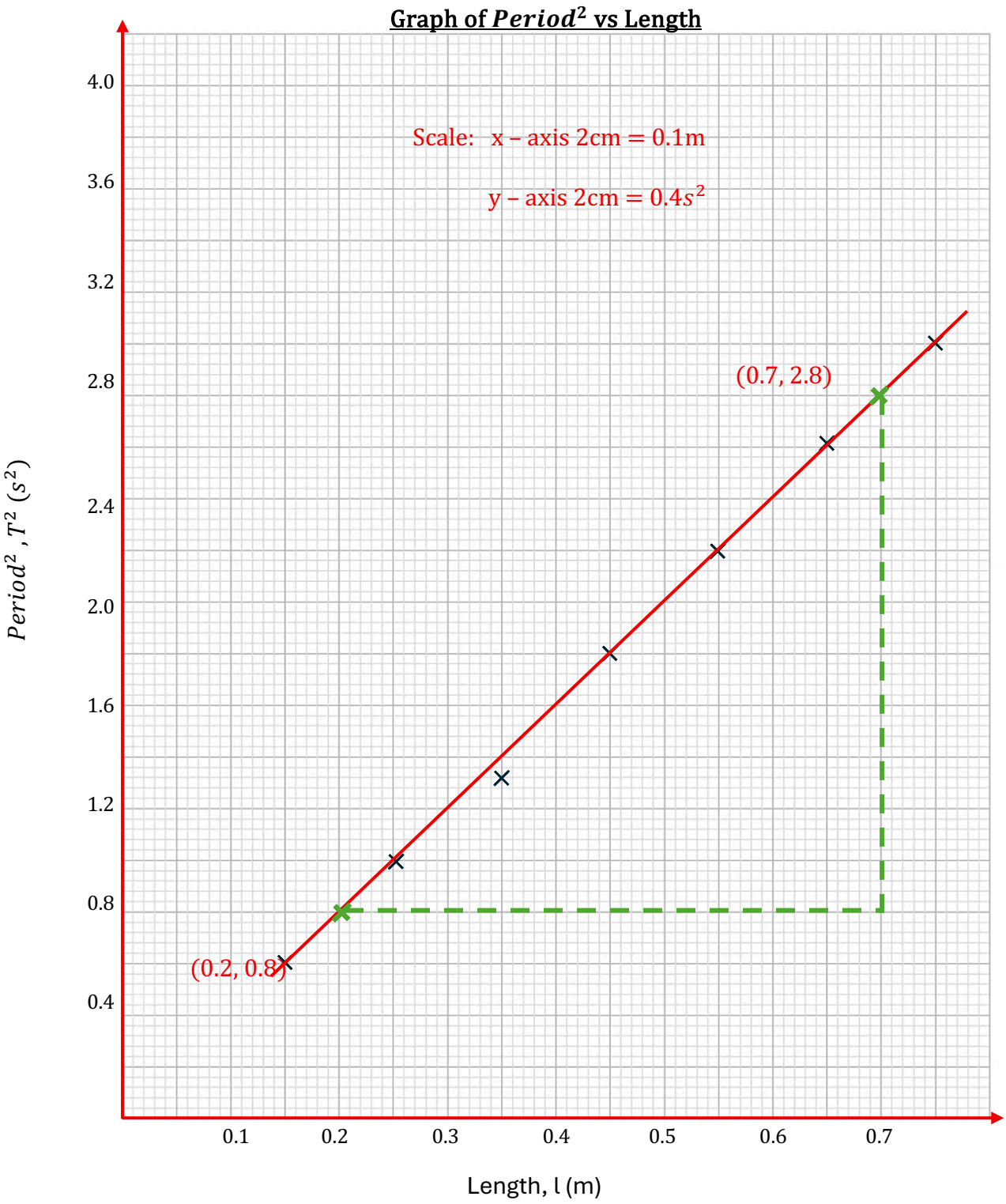
(iii) Using the grid provided in Figure 1 , plot a graph of the $Period^2$, T^2 versus length, l. (8 marks)



(c) Calculate the gradient, S, of the graph. (5 marks)

$$Gradient = \frac{y_2 - y_1}{x_2 - x_1}$$

Using points (0.2, 0.8) and (0.7, 2.8) from Graph

$$Gradient = \frac{2.8-0.8}{0.7-0.2}$$

$$= 4s^2m^{-1}$$

(d) (i) Given that $S = 4\pi^2/g$, calculate the acceleration due to gravity, g.

(4 marks)

$$S = \frac{4\pi^2}{g}$$

$$g = \frac{4\pi^2}{S}$$

$$= \frac{4\pi^2}{4}$$

$$= \pi^2$$

$$g = 9.87\ N/kg\ to\ 2\ dp$$

Total 25 marks

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2. (a) Complete the following statements by filling in the blanks with the correct word or phrase.

(i) The formula for pressure is given by _____ $P = \frac{F}{A}$ _____. Its SI unit is _____Pa_____ or _____ Nm^{-2}_____.

(ii) In fluids, pressure at all points on the same horizontal level is _____equal_____ and pressure increases as the _____depth_____ increases.

(iii) An object will float in a fluid when the _____ upthrust _____ on the object is equal to the _____ weight _____ of the object.

(7 marks)

(b) Figure 2 shows a shipwreck which is fully submerged at the bottom of the Caribbean Sea. The liquid pressure on the top of the surface of the shipwreck is 430 kPa.

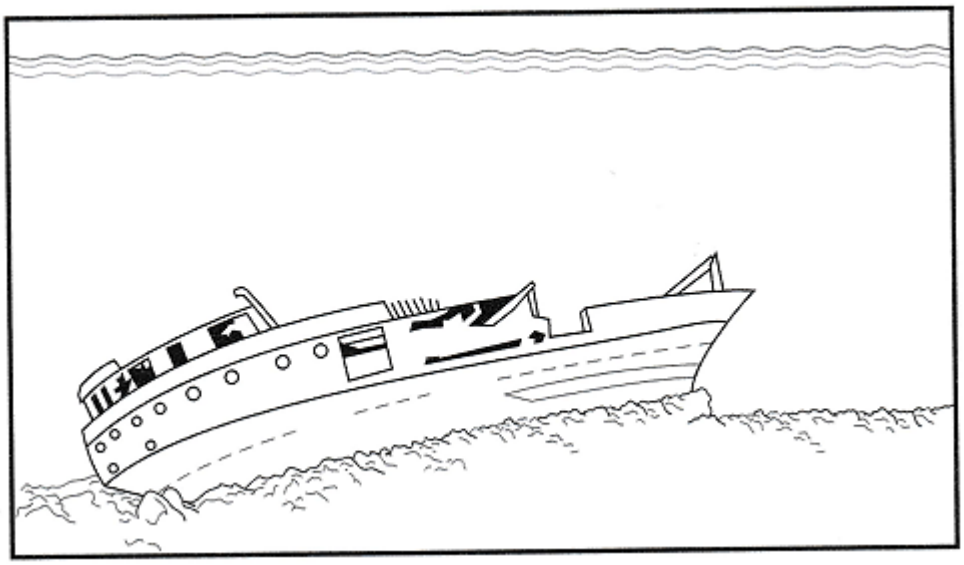

Figure 2. Diagram of a shipwreck

(i) Calculate the depth of the shipwreck below the surface of the sea, given that $g = 10Nkg^{-1}$, atmospheric pressure is 100kPa and the density of the sea water is 1025 kgm^{-3}. (4 marks)

$$P_l = \rho g h$$

$$h = \frac{P_l}{\rho g}$$

$$= \frac{430000}{1025 \times 10}$$

$$= 41.95m$$

(ii) Calculate the total pressure on the top surface of the shipwreck.

$$P_T = P_{atm} + P_l$$

$$= 100\,000 + 430\,000$$

$$= 530\,000 \text{ Pa}$$

(iii) A hatch door on the top surface of the shipwreck has an area of $0.60m^2$. Calculate the downward force on the hatch door due to the total pressure on the top surface of the shipwreck. (3 marks)

$$P_T = \frac{F}{A}$$

$$F = P_T \times A$$

$$= 530000 \times 0.6$$

$$= 318000\ N$$

Total 15 marks

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3. Figure 3 shows the section of a white plastic pipe through which water is flowing. The water in the pipe is heated by the sun .

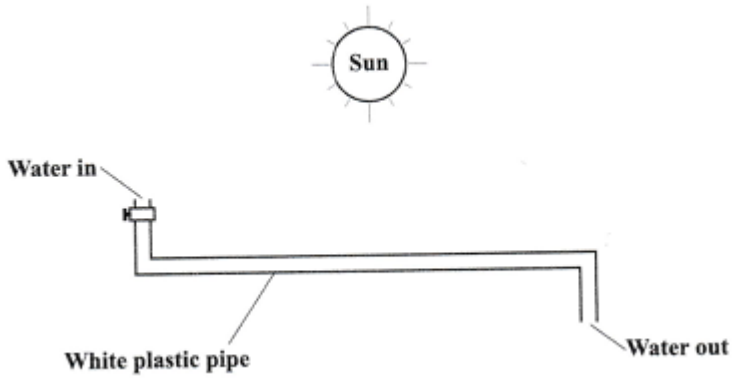

Figure 3. Device for heating water

(a) (i) Describe how thermal energy is transferred from the sun to the water inside the pipe (2 marks)

- The heat from the sun warms (transfers thermal energy) the pipe by radiation. This thermal energy is then transferred to the water molecules in the pipe by conduction.

(ii) Suggest TWO ways in which the efficiency of the thermal energy absorbed by the water in the pipe may be increased. (2 marks)

- Using a material that is a better conductor (metal like copper) will increase the amount of heat energy absorbed from the sun and transferred to the water
- Using black pipe instead of white increases the amount of radiation absorbed by the pipe.

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- Using a clear insulating material to surround the pipe allows for the pipe to be warmed by radiation from the sun while preventing heat loss from the pipe back to the atmosphere
- Increase the surface area of the pipe exposed to the sun

(b) Rachelle used a metal spoon to stir a hot liquid in a cup as shown in Figure 4.

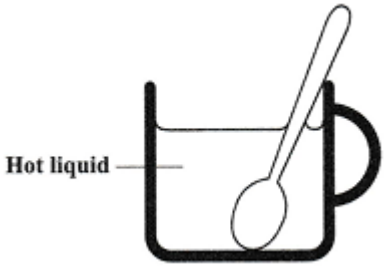

Figure 4. Metal spoon in a cup containing a hot liquid

(i) After a short time, Rachelle observed that the metal spoon felt hot. Describe the TWO ways in which thermal energy is conducted through the metal spoon at the molecular level. (3 marks)

- Energy is transferred to the metal atoms in contact with the water via conduction, increasing their vibrations in the lattice structure and the Kinetic energy at the molecular level is transferred molecule to molecule
- The free/delocalized electrons transfer energy on the molecular level in the form of kinetic energy from one to another and by moving.

(ii) The metal spoon was replaced by a wooden spoon. Rachelle observed that the wooden spoon did not feel hot. Explain why the wooden spoon did not feel hot. (2 marks)

- Wood is a poor conductor. It is a non-metal with no mobile electrons, therefore rate of heat transfer is reduced.

(c) A thermometer is placed in a plastic cup containing 200 g of water and gives a reading of 22 ^{0}C. Small pieces of ice at 0^{0}C are added to the water one by one. The mixture is stirred after each addition until the ice melts. The process was repeated until the temperature recorded by the thermometer was 0 ^{0}C. The total mass of ice added to the water was 60 g.

(i) Calculate the thermal energy lost by the original volume of water in the beaker, given that the specific heat capacity of water is $4.2Jg^{-1}{}^{o}\mathrm{C}^{-1}$.

Mass , m, = 200g

Initial Temp = $T_i = 22^0\mathrm{C}$

Final Temp = $T_f = 0^0\mathrm{C}$

Specific heat capacity of water $c = 4.2Jg^{-1}{}^{o}\mathrm{C}^{-1}$

$E_H = mc\Delta\theta$

$E_H = (200)(4.2)(22)$

$= 18480J$

(ii) Assume that all the thermal energy lost by the original volume of water in the beaker is transferred to the ice. Calculate the specific latent heat of fusion of ice.

Energy gained by ice = Energy lost by liquid Water

$Q = mL_f$

$L_f = \frac{Q}{m}$

$L_f = \frac{18480}{60}$

$= 308Jg^{-1}$

(3 marks)

Total 15 marks

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SECTION B

Answer ALL questions.

4. (a) Name the electromagnetic waves in EACH of the regions labelled A, B and C in the electromagnetic spectrum shown in Figure 5.

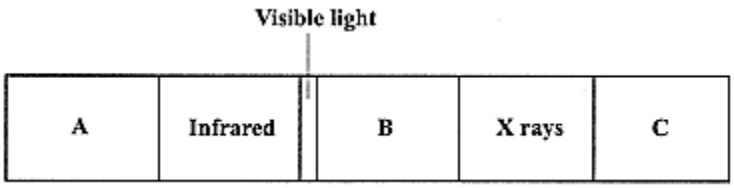

Figure 5. Electromagnetic Spectrum

A: Microwave

B:UltraVviolet (UV)

C: Gamma

(3 marks)

(b) Figure 6 represents some wavefronts approaching a barrier with a narrow gap, which is smaller than the wavelength of the wave. On Figure 6, draw THREE wavefronts after they have passed through the narrow gap.

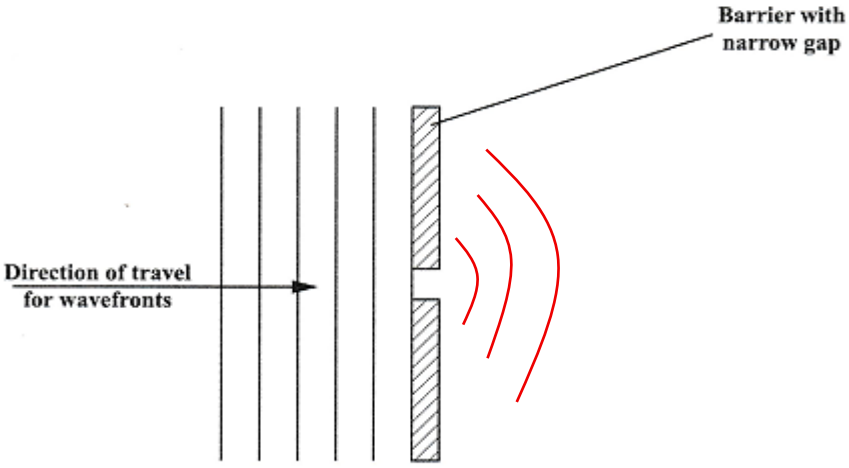

Figure 6. Wavefronts approaching a barrier with a narrow gap.

(c) Define the term 'refractive index of glass' (3 marks)

The refractive index of glass is the ratio of the speed of light in a vacuum (or air) to the speed of light in glass. It can also be expressed as the ratio of the sine of the angle of incidence in air to the sine of the angle of refraction in glass.

$$_{a}n_{g} = \frac{\sin i}{\sin r}$$

(d) Figure 7 shows a ray of red light incident on the face XY of a glass prism at point S. The path of the ray is shown in the diagram. The refractive index, n, of the glass for red light is 1.5.

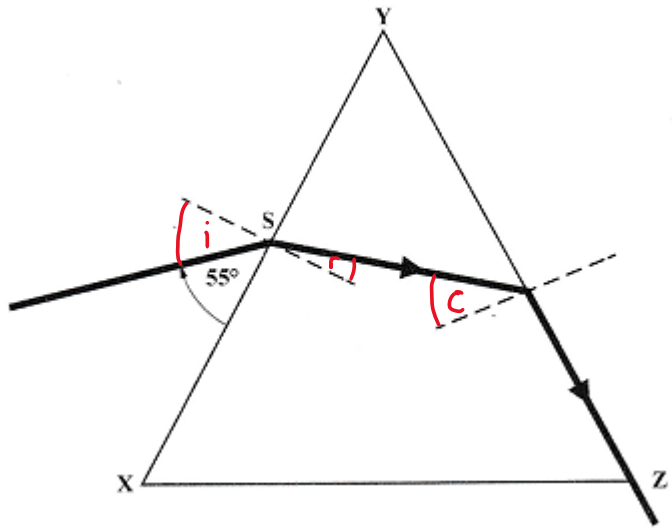

Figure 7. Ray of red light passing through a glass prism

(i) Calculate the angle of refraction in the glass at S. (3 marks)

$_an_g = 1.5$

$i = 90 - 55 = 35°$

$_an_g = \frac{\sin i\ (air)}{\sin r\ (glass)}$

$\sin r = \frac{\sin i}{n}$

$\sin r = \frac{\sin(35)}{1.5}$

$r = \sin^{-1}\left(\frac{\sin(35)}{1.5}\right)$

$r = 22.5^0$

(ii) Calculate the critical angle, c, for glass to air (3 marks)

$\sin c = \frac{1}{_an_g}$

$c = \sin^{-1}\left(\frac{1}{1.5}\right)$

$c = 42°$

(iii) Identify the critical angle, c, in Figure 7. (1 marks)

See Diagram critical angle , c.

Total 15 marks

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5. (a) Define the term 'electric field' (2 marks)

An electric field is the region around a charged object in which another charged object will experience a force

(b) (i) Figure 8 shows a positively charged sphere which is attached to an insulated stand. Draw arrows to indicate the pattern and the direction of the electric field in the region surrounding the sphere.

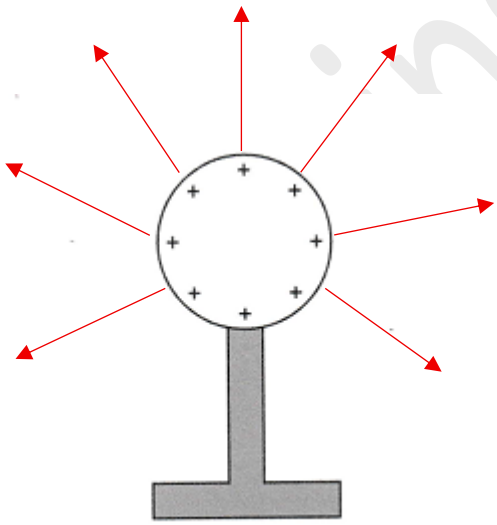

Figure 8. Positively charged sphere

(2 marks)

(ii) A smaller, uncharged metal sphere, Q, is suspended by a cotton thread and brought close to the positively charged sphere as shown in Figure 9. Draw the distribution of charge on Sphere Q (2 marks)

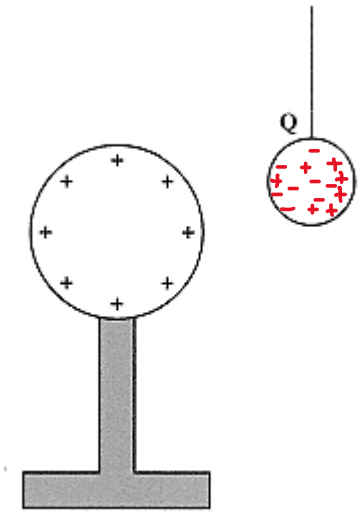

Figure 9. Positively charged sphere and metal sphere, Q.

(iii) An earth wire was briefly brought into contact with Sphere Q, as shown in Figure 10, and then removed.

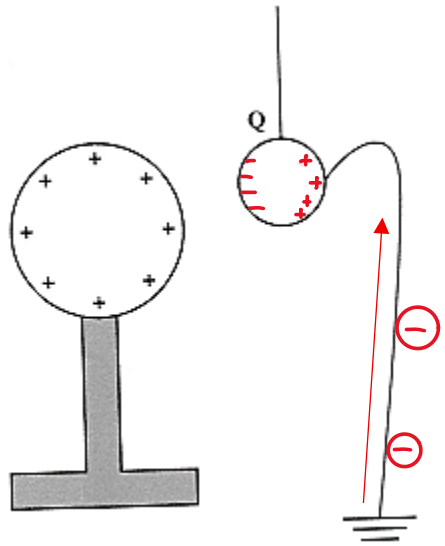

Figure 10. Positively charged sphere and metal sphere, Q, with earth wire

Describe what happens in the wire and state the final charge on Q.

Electrons flow from the earth along the ground wire towards the positively induced side of Sphere Q. These electrons then neutralise the positively induced side of the sphere. When the wire is removed, the net charge on the sphere is negative. (2 marks)

(d) (i) The battery in an electric car can be charged by passing a current of 11A through it. Calculate the charge stored in the battery after 8 hours.

(4 marks)

I = 11A

Time = 8 hours

$= 8 \times 60 \times 60\ seconds$

$= 28800s$

$Q = IT$

$Q = 11 \times 28800$

$Q = 316,800\ C$

(ii) Calculate the resistance of the battery given that the voltage used is 220V (3 marks)

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$$V = IR$$

$$R = \frac{V}{I}$$

$$R = \frac{220}{11}$$

$$= 20\Omega$$

Total 15 marks

6. (a) Complete 2 by inserting the types of radioactive emissions and their properties.

TABLE 2: RADIOACTIVE EMISSIONS AND PROPERTIES

Radioactive Emissions	Properties		
	Nature	Range in Air	Stopped By
Gamma	Electromagnetic radiation	Several kilometres	Several inches of lead
Alpha	Helium nucleus	1-3cm	1 sheet of paper
Beta	Electron	A few metres	A few mm of aluminium

(3 marks)

(b) Describe the term 'isotopes' (3 marks)

Isotopes are atoms of the same element, having the same atomic number, number of protons but having a different number of neutrons, thus having different mass numbers. Eg C-12 and C-14

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(c) The structure of the atom of an element Y, showing the number of neutrons (n), protons(p) and electrons €, is represented in Figure 11.

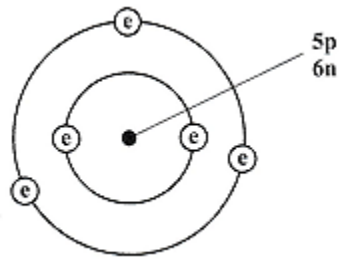

Figure 11. Structure of atom of Element Y.

(i) State the mass number and atomic number of this atom.

Mass Number : 11

Atomic Number: 5

(2 marks)

(ii) Write the symbol for another isotope of Element Y. (1 mark)

$^{12}_{5}Y$

(d) (i) A polonium -210 nucleus decays by α-particle emission to a lead (Pb) nucleus. Complete the equation for the decay of polonium – 210.

$$^{210}_{84}Po \rightarrow \ ^{206}_{82}Pb + \ ^{4}_{2}He$$

(ii) Polonium 210 has a half life of 138 days. At a certain time, a sample contains 6.4 $\times 10^6$ polonium nuclei. Calculate the number of polonium nuclei remaining after 552 days.

$$\#\ of\ half\ lives = \frac{552}{138}$$

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$$= 4\ half\ lives$$

$$After\ 4\ half\ lives, remaining\ fraction = \left(\frac{1}{2}\right)^4$$

$$After\ 4\ half\ lives, remaining\ fraction = \left(\frac{1}{16}\right)$$

$$\text{Number of remaining Sample} = \left(\frac{1}{16}\right) \times 6.4 \times 10^6$$

$$= 4.0 \times 10^5\ Polonium\ Nuclei$$

(3 marks)

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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